

K-MEANS

$$\min_c \sum_i \min_k d(X_i, C_k)$$

$\Leftrightarrow$

$$\min_{c, M} \sum_{i, k} M_{ik} \cdot d(X_i, C_k)$$

$$\sum_k M_{ik} = 1, \forall i$$

$$M_{ik} \in \{0, 1\}$$

$$X = \begin{matrix} & 1 & j & M \\ \begin{matrix} 1 \\ \vdots \\ i \\ \vdots \\ N \end{matrix} & \begin{bmatrix} \\ \\ \\ \end{bmatrix} \end{matrix}$$

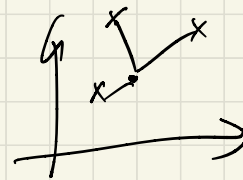
$$C = \begin{matrix} & 1 & j & M \\ \begin{matrix} 1 \\ \vdots \\ k \\ \vdots \\ K \end{matrix} & \begin{bmatrix} \\ \\ \\ \end{bmatrix} \end{matrix}$$

PROBLEM: j

$$M = \begin{matrix} & 1 & K & K \\ \begin{matrix} 1 \\ \vdots \\ i \\ \vdots \\ N \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

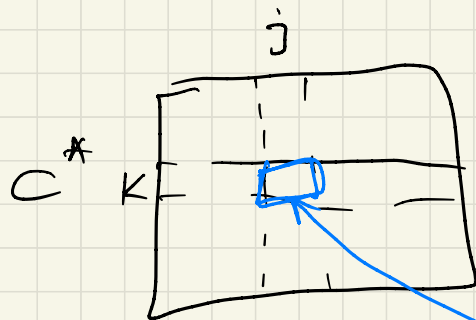
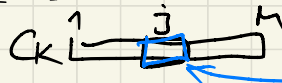
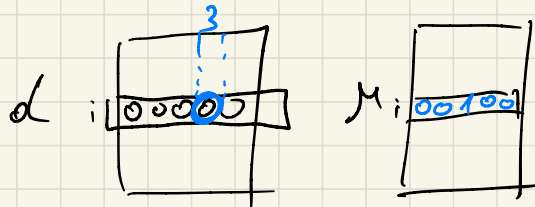
$$d = \begin{matrix} & 1 & K & K \\ \begin{matrix} 1 \\ \vdots \\ i \\ \vdots \\ N \end{matrix} & \begin{bmatrix} 0,2 & 0,8 & 1 & 4 \end{bmatrix} \end{matrix}$$

$$d(x, y) = \sqrt{\sum_j (x_j - y_j)^2}$$



$$1^{\circ} \min_{i,k} \sum_{i,k} M_{ik} \circ d_{ik} \quad \text{CONST } C - \text{CONST}$$

$$M_i = \underset{K}{\text{ARGMIN}} \circ d_{ik}$$



BR INST.
KOJI
PRIPADU
K

$$2^{\circ} \min_C \sum_K \sum_i M_{ik} \cdot d(X_i, C_k)$$

$$\forall_K \min_{C_K} \sum_i M_{ik} \cdot d(X_i, C_k)$$

$$\min_{C_K} \sum_{i: M_{ik}=1} \sum_j (X_{ij} - C_{Kj})^2 = E$$

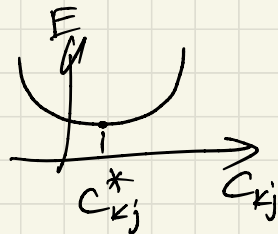
$$\frac{\partial E}{\partial C_{Kj}} = \sum_{i: M_{ik}=1} 2 \cdot (X_{ij} - C_{Kj}) \cdot (-1) = 0$$

$$\begin{aligned} \sum_{i=1}^{N_K} (X_{ij} - C_{Kj}) &= 0 \\ \sum_{i=1}^{N_K} X_{ij} - \sum_{i=1}^{N_K} C_{Kj} &= 0 \\ \sum_{i=1}^{N_K} X_{ij} &= \sum_{i=1}^{N_K} C_{Kj} \\ \sum_{i=1}^{N_K} X_{ij} &= N_K \cdot C_{Kj} \end{aligned}$$

$$C_{Kj}^* = \frac{1}{N_K} \sum_{i=1}^{N_K} X_{ij}$$

AVG
MEAN

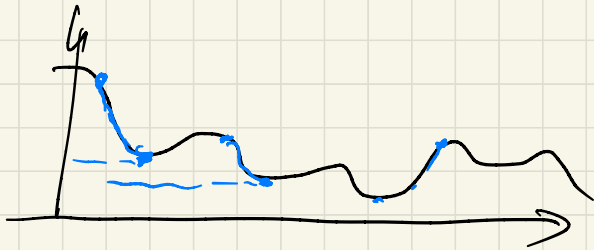
BLOCK
COORDINATE
DESCENT



ALGORITAM

K - FIX

1. INIT C - RANDOM (BOUNDS?)
2. CALC. $\mu_i \forall i \Rightarrow$ FIX μ
3. CALC. $C_k^* \forall k \Rightarrow$ FIX C
4. IF C_k PROMENJENO: REPEAT FROM 2.



NAPOMENE:

- F-ja cilja nije KONVEKSIJA!
COMPACTNESS, SSE
- REŠENJE ZAVISI OD INIT
- VRLO BRZO IZVRŠAVANJE
- α je SUMA PO j
(ZAVISI OD SKALE ATRIBUTA)
- RAZL. KLASITERI MOGU IMATI
RAZLIČITE SSE
KVALITETE

